

BANK DIGITAL TWIN SIMULATION PLATFORM

1. Executive Summary & Strategic Business Rationale

1.1 The Concept of a Financial Digital Twin

A Bank Digital Twin is a software replica of a commercial banking institution. In physical industries such as aerospace and automotive engineering, digital twins simulate jet engines and complex supply chains to detect failures before they occur in the physical world. Similarly, the Bank Digital Twin simulates a bank's entire operational ecosystem—including synthetic customer pools, deposit accounts, retail and commercial loan portfolios, macroeconomic environment variables, and real-time transaction streams.

By mapping accounting balance sheet relationships and statistical risk models into an integrated software platform, financial executives can test critical business hypotheses without placing real balance sheet capital or customer deposits at risk.

1.2 The Core Problem Solved

Modern commercial banks operate under strict regulatory frameworks (Basel III/IV, CCAR, DFAST) and face unprecedented macroeconomic volatility. Traditional risk assessment methods suffer from three systemic limitations:

- **High Latency in Stress Evaluation:** Running full Monte Carlo portfolio stress tests traditionally requires hours of mainframe batch processing, preventing real-time decision-making during fast-moving economic crises.
- **Siloed Risk Analytics:** Credit risk, market liquidity risk, fraudulent transaction monitoring, and regulatory compliance reporting are typically managed in separate, fragmented software tools.
- **Black-Box Machine Learning Models:** Legacy machine learning algorithms produce risk predictions without clear feature attributions, making it difficult for compliance officers and bank examiners to audit model outputs.

1.3 Strategic Business Value & Solvency Defense

The Bank Digital Twin addresses these limitations by establishing a single, unified simulation environment. Bank leadership, Chief Risk Officers (CROs), quantitative analysts, and regulatory auditors gain immediate visibility into:

- Instant Solvency Assessment under severe macroeconomic shocks (e.g., commercial real estate crashes, stagflation, aggressive interest rate hikes).
- Real-Time What-If Hypothesis Testing yielding sub-50-millisecond balance sheet recalculations for rapid executive decision support.
- Multi-Horizon Predictive Forecasting for capital adequacy ratios, non-performing loan proportions, and liquidity ratios with 95% statistical confidence bounds.
- Unsupervised Transaction Anomaly Detection to intercept high-risk wire transfers, crypto velocity surges, and money laundering vectors in real time.

2. Comprehensive Financial & Technical Glossary

To ensure complete clarity across both executive leadership and technical teams, key financial metrics, regulatory standards, and quantitative methodologies utilized throughout the platform are formally defined below:

Term / Metric	Formal Definition & Business Significance
Capital Adequacy Ratio (CAR %)	The ratio of a bank's core capital to its risk-weighted assets. It measures a bank's financial strength and capacity to absorb operational losses. Under Basel III, banks must maintain a minimum CAR of 8.0% (plus a 2.5% Capital Conservation Buffer).

Non-Performing Loans (NPL %)	The percentage of total loans in default or close to default (typically 90+ days past due). High NPL ratios indicate deteriorating borrower credit quality and increased provision requirements.
Risk-Weighted Assets (RWA)	The total asset value held by a bank, weighted by regulatory loss coefficients according to risk level (e.g., cash carries 0% weight, while commercial mortgages carry higher weightings).
Liquidity Coverage Ratio (LCR %)	The proportion of high-quality liquid assets (HQLA) held by a financial institution to ensure its ongoing ability to meet short-term liquidity obligations during a 30-day severe stress period (regulatory minimum is 100%).
Net Stable Funding Ratio (NSFR %)	The ratio of available stable funding relative to required stable funding over a one-year horizon, ensuring banks maintain a stable funding structure for long-term assets.
Vasicek Credit Model	A single-factor Gaussian correlation framework used in quantitative finance to model portfolio credit losses under macroeconomic stress by linking individual borrower default rates to a common systemic economic factor.
Value at Risk (VaR 99%)	A statistical measure indicating the maximum estimated monetary loss expected across a loan portfolio over a specified timeframe at a 99% confidence level.
Expected Shortfall (CVaR 99%)	The conditional expectation of loss magnitude beyond the 99% VaR threshold, quantifying the severity of extreme tail losses during severe systemic crises.
IsolationForest Algorithm	An unsupervised machine learning decision-tree algorithm that isolates anomalous data points based on feature space partitioning. It scores multivariate transaction outliers without requiring pre-labeled training datasets.
SHAP (SHapley Additive exPlanations)	A game-theoretic framework that explains individual machine learning model predictions by calculating the exact marginal contribution of each borrower attribute (e.g., credit score, income, debt-to-income).

3. System Architecture & Technology Stack

3.1 Architectural Principles

The platform is engineered using a decoupled, multi-tier micro-service architecture. The system separates frontend user interaction, high-speed asynchronous application programming interface (API) routing, specialized computational math engines, and persistence layers.

Key architectural features include:

- **Asynchronous Non-Blocking Processing:** Built on FastAPI and Uvicorn to deliver concurrent handling of quantitative calculations and API requests.
- **Stateless Computational Engines:** Mathematical simulation services operate independently, enabling horizontal scaling under high workload concurrency.
- **Dual Database Compatibility:** Native SQLAlchemy 2.0 object-relational mapping supports zero-setup local SQLite storage as well as enterprise PostgreSQL instances.

3.2 Technical Component Rationale

System Layer	Technology Choice	Architectural Rationale
Frontend Framework	Next.js 16 (App Router) & React 19	Provides hybrid server/client rendering, fast route transitions, and structured component architecture.
UI & Styling	Tailwind CSS & Plus Jakarta Sans	Delivers high-density dark glassmorphic layouts, consistent typography, and clean corporate aesthetics.
Data Visualization	Recharts & Custom 3D SVG Canvas	Enables interactive financial area charts, sensitivity tornado bars, and custom isometric 3D pie charts.
Backend ASGI Server	FastAPI & Uvicorn (Python 3.11)	Delivers high-speed asynchronous routing with automated OpenAPI documentation and schema validation.
Quantitative Math	NumPy 2.3 & SciPy 1.17	Vectorized matrix operations supporting 10,000 Monte Carlo simulation paths in under 500 milliseconds.
Machine Learning	Scikit-Learn 1.9 & SHAP 0.51	Powers calibrated Random Forest default risk scoring, IsolationForest anomaly detection, and SHAP explainability.
Time-Series Forecasting	Statsmodels 0.15	Computes Exponential Smoothing and ARIMA models with 80% and 95% statistical confidence bounds.
Persistence & ORM	SQLAlchemy 2.0 & SQLite / PostgreSQL	Relational database mapping supporting zero-configuration SQLite for local use and PostgreSQL for production.

4. Machine Learning & Quantitative Modeling Deep-Dive

4.1 Vectorized Vasicek Monte Carlo Stress Engine

The core quantitative stress testing engine utilizes the Vasicek Single-Factor Credit Loss Model. In this framework, the asset value of borrower i is governed by a systemic macroeconomic factor Z and an idiosyncratic risk factor E_i :

$$A_i = \sqrt{\rho} * Z + \sqrt{1 - \rho} * E_i$$

Where ρ represents the asset correlation among portfolio obligors. When severe macroeconomic shocks depress Z , borrower asset values drop simultaneously, causing systemic default correlation spikes. The engine vectorizes matrix operations across 10,000 simulated capital paths over 60-month evaluation windows, calculating P5, P50, and P95 drawdown trajectories in <500ms.

4.2 Surrogate Response Surface Engine (<50ms What-If Evaluation)

To enable real-time interactive hypothesis testing without requiring full Monte Carlo re-execution on every slider adjustment, the platform utilizes a pre-trained polynomial surrogate response surface. The surrogate model maps four macroeconomic shock variables:

- Unemployment Rate Shock (3.5% to 20.0%)
- Interest Rate Hike (-2.0% to +8.0%)

- Commercial Property Price Drop (0.0% to 50.0%)
- Deposit Outflow Percentage (0.0% to 40.0%)

The surrogate model evaluates post-stress Capital Adequacy Ratio (CAR %), Non-Performing Loan Ratio (NPL %), and Net Loss figures in <50ms. Simultaneously, the engine calculates the partial derivatives of CAR % with respect to each macro shock, dynamically driving the Sensitivity Tornado Ranking Chart to highlight the primary risk driver.

4.3 Unsupervised IsolationForest Fraud Sentinel

Transaction monitoring combines rule-based heuristic filters with an unsupervised IsolationForest machine learning model. The model builds an ensemble of isolation trees over feature vectors comprising transaction amount, time-of-day, historical velocity, and category risk index.

Anomalous transactions are isolated near the root of decision trees, producing an ML anomaly score (0.0 to 1.0). This score is combined with rule-based triggers (e.g., Large Wires > \$10k, Midnight Activity, Velocity Spikes) to assign risk tiers (Critical, High, Medium, Low) and calculate aggregate capital exposure.

4.4 Statsmodels Time-Series Forecasting Engine

Financial metric projections (CAR %, NPL %, NIM %, Liquidity Ratio %, Tier 1 Capital) are calculated using Holt-Winters Exponential Smoothing and Autoregressive Integrated Moving Average (ARIMA) models. The engine computes point forecasts alongside upper and lower 95% statistical confidence bounds over 30-day, 60-day, and 90-day horizons.

4.5 Calibrated Credit Risk Scorecard & SHAP Explainability

Individual borrower risk scoring utilizes a Random Forest classifier calibrated via Isotonic Regression. To ensure complete auditability, the platform integrates SHAP (SHapley Additive exPlanations) TreeExplainer, quantifying the exact impact of income, credit score, debt-to-income, and employment status on default probability.

5. Comprehensive Functional Module Breakdown

The Bank Digital Twin consists of nine fully integrated operational modules accessible via the navigation interface:

1. Overview Dashboard (/)

Displays baseline bank solvency KPIs (CAR %, NPL %, Tier 1 Capital, RWA, ROA, NIM). Features 3D flippy hover cards, 30-day historical trend graphs, asset class risk breakdowns, structured JSON/CSV data export options, and a collapsible AI Chief Risk Officer executive memo.

2. What-If Interactive Simulator (/what-if)

Provides real-time macroeconomic shock sliders. Instantly calculates post-stress CAR %, NPL %, net loss estimates, and solvency risk levels in <50ms. Displays a dynamic Sensitivity Tornado Ranking Chart ranking shock factors by marginal CAR impact.

3. Predictive Metric Forecasting (/forecasts)

Offers multi-metric time-series projections (CAR %, NPL %, NIM %, Liquidity, Tier 1) across 30, 60, and 90-day horizons with 95% confidence intervals. Features side-by-side equal-height cards, risk zone pie distributions, and an isometric 3D cylinder pie chart (ThreeDPieChart).

4. Fraud & Anomaly Intelligence Center (/fraud)

Monitors transaction streams using IsolationForest ML and multi-vector heuristic rules (large wires, midnight transfers, high velocity). Displays total alerts, capital exposure (\$), ML precision metrics, risk tier donut charts, live threat simulation triggers, and searchable flagged transaction tables.

5. Monte Carlo Stress Testing Engine (/stress)

Executes 10,000-path Vasicek simulation runs based on historical or custom stress parameters. Generates drawdown fan charts (P5/P50/P95), loss severity histograms, Value at Risk (VaR 99%), and Expected Shortfall (CVaR 99%) metrics.

6. Loan Portfolio Explorer (/portfolio)

Searchable catalog of commercial and retail loans with risk status filters. Integrates an Explainable AI Credit Risk Scorecard modal displaying individual borrower default probabilities and SHAP feature attributions.

7. Stress Scenario Library (/scenarios)

Catalog of pre-configured macroeconomic stress scenarios (2008 Financial Crisis, 2023 SVB Liquidity Run, Commercial Real Estate Collapse, Stagflation Shock) and natural language scenario synthesis capabilities.

8. Regulatory Compliance & Reporting (/compliance & /reports)

Automated compliance matrix evaluating balance sheet metrics against Basel III / IV standards (CAR $\geq$ 8%, Tier 1 $\geq$ 6%, LCR $\geq$ 100%, NSFR $\geq$ 100%). Generates downloadable PDF, CSV, and JSON audit reports.

9. Execution Audit History (/history)

Audit trail table logging all executed stress tests, user timestamps, input parameters, and historical capital impact results.

6. Quality Assurance, Testing & Performance Benchmarks

To ensure enterprise reliability, financial accuracy, and software stability, the platform has undergone rigorous automated testing and static analysis:

- **100% Pytest Backend Test Pass Rate:** All 16 unit and integration test suites pass with zero failures across auth, compliance, forecasting, fraud detection, KPI calculation, ML risk scoring, simulator math, transaction processing, and What-If surrogate models.
- **Zero TypeScript Compilation Errors:** The Next.js 16 frontend codebase compiles with zero type errors (verified via `npx tsc --noEmit`), guaranteeing strong client-side type safety.
- **Sub-50ms Response Latency:** Surrogate model What-If evaluations execute in <50 milliseconds, enabling smooth, uninterrupted slider interactions.
- **High-Throughput Monte Carlo Execution:** Vectorized matrix operations allow 10,000 simulation paths across 60-month horizons to complete in under 500 milliseconds.

7. Step-by-Step Client Demonstration Workflow

When presenting the Bank Digital Twin to executive stakeholders, risk committees, or regulatory auditors, the following five-step demonstration workflow is recommended:

Step 1: Baseline Solvency & Balance Sheet Review (Overview Dashboard)

Begin on the Overview Dashboard (<http://localhost:3000/>). Present the bank's baseline Capital Adequacy Ratio (CAR: 15.2%) and NPL Ratio (2.5%). Hover over the KPI cards to demonstrate the 3D tilt effects, and scroll below the fold to highlight the collapsible AI CRO Copilot memo and Basel III compliance badge.

Step 2: Real-Time Macroeconomic Hypothesis Testing (What-If Simulator)

Navigate to the What-If Simulator (<http://localhost:3000/what-if>). Explain: 'Instead of waiting hours for batch simulations, our surrogate model recalculates balance sheet impacts instantly.' Adjust Unemployment to 15.0% and Property Drop to 25.0%. Point out how post-stress CAR % and Net Loss update in <50ms and show the Sensitivity Tornado Ranking Chart automatically re-ordering risk factors.

Step 3: Multi-Horizon Metric Forecasting (Predictive Forecasting)

Navigate to Predictive Metric Forecasting (<http://localhost:3000/forecasts>). Select Capital Adequacy Ratio (CAR %) and set the horizon to 90 Days. Highlight the 95% statistical confidence bounds on the trend area chart and showcase the Isometric 3D Cylinder Pie Chart displaying risk zone distribution slices.

Step 4: Live Transaction Anomaly Interception (Fraud Center)

Navigate to the Fraud & Anomaly Intelligence Center (<http://localhost:3000/fraud>). Review the IsolationForest ML precision metrics and active sentinel alert feed. Click 'Simulate Anomaly Threat' to trigger a live synthetic crypto/wire threat and demonstrate real-time interception with toast notification feedback.

Step 5: Portfolio Analysis & Regulatory Audit (Stress Engine & Reports)

Navigate to the Monte Carlo Stress Engine (<http://localhost:3000/stress>). Select the '2008 Financial Crisis' preset scenario and click 'Run Stress Test' to execute 10,000 Vasicek simulation paths. Review the drawdown fan chart, VaR 99%, and Expected Shortfall, then export the regulatory compliance summary.

8. Conclusion & Strategic Roadmap

The Bank Digital Twin represents a significant advancement in enterprise financial risk intelligence. By uniting quantitative Monte Carlo stress testing, real-time What-If surrogate modeling, time-series forecasting, unsupervised machine learning fraud sentinel streams, and automated Basel III regulatory compliance into a clean, modern interface, the platform provides financial institutions with unprecedented decision velocity and capital protection.

Future expansion capabilities supported by the architecture include:

- Real-Time Core Banking Integration: Direct connection to live core banking ledgers via SWIFT and ISO 20022 message streams.
- Automated Regulatory Reporting Pipelines: Direct XML/XBRL export formatting for Federal Reserve and European Central Bank regulatory filings.
- Deep Reinforcement Learning Capital Allocation: Autonomous portfolio rebalancing models optimizing Risk-Weighted Assets against target Return on Equity.